Refining Hardness after ETH

Task-Relevant Distinctions and the Cost of Exact Counting

Rohan Vinaik

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Abstract

The reported subexponential algorithms for satisfiability leave a strong successor question: can exact counting gain a fixed power in its exponent? We examine that question through the distinctions a task requires, the operations that resolve them jointly, and what those operations cost, with the public hypotheses #SH and Weak #ETH as the quantitative setting. After comparing the existing method-specific barriers, we give a fully formalized finite analysis of the coordinate-swap construction behind one interpolation algorithm. We construct the measurement rows it selects, prove their fixing-kernel property and their exact row count, and connect them to a uniform trade-off among measurement, degree and enumeration. Under the stated row and dimension budgets, the complete clipped substitution enumeration has size at least $2^{n \log_2 \log_2 n/(4\sqrt{\log_2 n})}$ for $n \geq 2^{16}$, and eventually at least $2^{n^{1-\varepsilon}}$ for every fixed $\varepsilon > 0$. This bounds the enumeration as completed; an algorithm that evaluates its aggregate by other operations calls for a separate analysis. The supporting mathematical implications and the constructed-measurement application are checked in Lean 4, and the unrestricted counting hypothesis stays open.

1 Introduction

The Exponential Time Hypothesis fixes the difficulty of 3-SAT at one scale: no deterministic algorithm runs in time $2^{o(n)}$, up to polynomial factors in the input, where n counts the variables [9]. Two recent preprints report algorithms that cross this boundary [7, 13]. Both algorithms count satisfying assignments exactly rather than merely deciding satisfiability. If their analyses hold, ETH is false. What they leave open is how much further the exponent can fall.

That gap is both quantitative and structural. A running time of $2^{n/\sqrt{\log n}}$ is subexponential, yet for every fixed $\varepsilon > 0$ its exponent still exceeds $n^{1-\varepsilon}$ once n is large. A conjecture that rules out the latter scale therefore survives such an algorithm untouched. The Satisfiability Hypothesis of Stearns and Hunt [14], with its counting analogue, asks exactly this successor question. MIT Hardness Group [11] have since shown it equivalent to Weak ETH and Weak #ETH, and they have built the matching reduction theory.

We ask what the distinction means for an account of hardness. A problem is hard neither because its presentation has many variables nor because its solution space has many candidates. One operation can resolve many apparent alternatives at once, while a compact representation of the answer can still be expensive to evaluate. So a refined hardness claim has to name four things together: the task, the alternatives that change its answer, the operations permitted on the input, and the cost of obtaining and using what those operations return.

Our thesis is that the failure of ETH's original scale calls for this more discriminating account, and not for abandoning strong hardness hypotheses. Exact counting is a sharp setting for it: its

answer is short, but computing that answer means controlling interactions among many assignments. Both reported algorithms show that those interactions can be organized more cheaply than ETH allowed. What remains is whether successive reorganizations can lower the exponent by a fixed power, or whether a near-exponential obstruction persists. We take #SH, already public, as the quantitative form of that question, and we introduce no new name for it.

Sections 3 and 4.1 mark two limits on what an information argument alone can prove. Section 3 gives an exponential point-query separation between counting and decision on small CNFs. Reading the formula as given costs only its encoding length in bit queries, so carrying that separation to running time would need a justified access simulation and its own account of computation. Section 4.1 gives the exact Hermite threshold for arbitrary degree-bounded polynomials. Its necessity proof uses indistinguishable polynomials that need not arise from formulas, and its threshold counts coefficients supplied rather than computed, so it says nothing yet about the cost of producing them.

Those two checks bound the lower-bound argument from the outside; Theorem 5.1 rests on neither. Section 4 looks at the work the algorithms do on the input as written. Section 5 then finds an obstruction inside the operations of one such construction. Its coordinate swaps have an exact spectrum, so the fixing condition imposes a rank requirement on every rational measurement map. Agreement with parity turns that requirement into a degree floor, and the degree and dimension budgets then force a near-exponential complete substitution enumeration. We also build the measurement rows that the source algorithm selects, and we prove that they satisfy the fixing property.

The spectral mechanism and the near-exponential template scale both have public antecedents, which Section 4.3 discusses. We add the finite, uniformly quantified implication itself, its mechanization on the coordinate space the algorithm uses, and the checked application to the source's own measurements. Together these give one concrete instance of the broader structural program: a result that says which changes of fixing polynomial or measurement rows yield no fixed-power improvement while the other contracts and the complete enumeration stay in place. Whether the unrestricted successor hypothesis holds is another question, and it stays open.

2 The quantitative question after ETH

2.1 Exponential scale and power of the exponent

Let L denote a structural encoding length of a CNF formula, counting its variables, clauses and literal occurrences. Conventional indexed binary encodings change this length by at most polylogarithmic factors, and such changes preserve the fixed-power separation in the hypotheses below. An arbitrary polynomial blowup needs its own accounting. Following the conventions of [11], we include unused variables in L, and the variable-parameter hypothesis keeps polynomial running-time factors in the full input length in view.

Hypothesis 2.1 (Weak #ETH and #SH). Weak #ETH asserts that for every fixed $\varepsilon > 0$, no deterministic algorithm computes the exact number of satisfying assignments of a 3-CNF formula in time

$$O\Big(2^{n^{1-\varepsilon}}\Big)\, L^{O(1)}.$$

The hypothesis #SH asserts that, for every fixed $\varepsilon > 0$, exact counting for unrestricted CNF formulas has no deterministic running time $O(2^{L^{1-\varepsilon}})$.

MIT Hardness Group prove that these two hypotheses are equivalent [11, Theorem 2.4]. Their counting-preserving ingredients are a linear-size Tseitin conversion with unique gate extensions and a disjoint, quantitative sparsification procedure, and their decision versions relate SH and Weak ETH in the same way. Both are public hypotheses with public reductions, so our unconditional theorems

assume neither.

This distinction from ETH is more than terminology. For any fixed $a > 0$ and fixed real b, a positive exponent of order

$$\frac{n(\log\log n)^b}{(\log n)^a}$$

is $o(n)$ and yet $n^{1-o(1)}$. An algorithm at such a scale therefore refutes ETH while remaining compatible with Weak #ETH. Calling it subexponential is correct; concluding that it lowers the exponent by a fixed power would be wrong.

Both source results sit precisely there. For each fixed clause width, OpenAI report a counting time on general formulas of that width of

$$L^{O(1)} \exp\Big(O\Big(\frac{n\log\log n}{\sqrt{\log n}}\Big)\Big)$$

and a sparse-formula bound with exponent $O(n/\sqrt{\log n})$ [13]. The Claude paper reports, for fixed clause width k, the bound

$$2^{O\left(\min\left\{\frac{km}{\sqrt{\log m}}, \frac{k^2 n\log\log n}{\sqrt{\log n}}\right\}\right)},$$

where m counts clauses and n counts the variables that occur [7]. Unused variables contribute a factor to the count that is computed directly. We take both as the upper bounds their authors state; the present formalization has not re-verified them.

2.2 What a structural explanation must add

#SH states a rate of growth, and nothing more. It leaves unexplained which aspect of the input prevents a faster computation. An account of that obstruction needs more than the number of possible assignments, because summation, elimination, symmetry and algebraic identities all treat many assignments together.

We therefore ask three questions. Which alternative instances require different task answers? Which permitted operations distinguish those alternatives? And how much computation goes into selecting, constructing, evaluating and combining those operations? Two of these fix an information requirement, while the third asks how an algorithm realizes it, and the analysis below needs all three.

This separation also clarifies what a change of representation can do. A new representation may preserve every task-relevant distinction while making their joint resolution far cheaper. It may instead introduce an operation that is substantially more informative than the old ones. A certificate bound transfers only when a stated simulation preserves the relevant access costs, and a running-time claim needs computational control of the representation and its decoder as well. So there is no general inference from preserving information to preserving computational difficulty.

Our structural reading of #SH is correspondingly specific. Can every attempt at a fixed-power improvement be shown to leave an expensive unresolved interaction, once all jointly computable operations have been charged? Posed for unrestricted algorithms that is a research question. Section 5 answers one instance of it inside a defined construction, with the operators the algorithm uses and with stated budgets. Section 6 sets out what still stands between that instance and an unrestricted result.

3 Distinctions required by a task

3.1 Certificates and completed computations

Let $\mathcal{C}$ be a class of admissible instances, let $q : \mathcal{C} \to Y$ be the task, and let each $e \in E$ specify an observation $o_e : \mathcal{C} \to A$. The answer type may encode several scalars or a structured object. The class $\mathcal{C}$ belongs to the model, since it says which alternatives a computation has to handle.

Definition 3.1 (Task certificate). A finite set $T \subseteq E$ is a certificate for an instance $x \in \mathcal{C}$ if

$$\forall y \in \mathcal{C}, \quad (\forall e \in T,\ o_e(y) = o_e(x)) \Longrightarrow q(y) = q(x).$$

Write $\mathrm{cert}_q(x)$ for its minimum cardinality, with value ∞ when no finite certificate exists. With nonnegative observation charges c_e, use the infimum of $\sum_{e \in T} c_e$ over finite certificates instead, again taking ∞ for an empty family.

This is the usual certificate notion for the declared observations, extended to a general task, and we claim no new complexity measure for algorithms in it: standard decision-tree and priced-information treatments already distinguish certificates, queries and the cost of finding useful information [5, Section 4.1; 6].

A deterministic observation policy chooses its next observation, or its final answer, from the previous query–answer history alone. Any further access to the instance therefore has to appear among the permitted observations.

Proposition 3.2 (Transcript certificate). *If a deterministic policy halts on x with the correct task answer and is correct whenever it halts on an admissible instance, its completed observation transcript is a certificate for x. In particular, the number of observation calls on a correct terminating run is at least* $\mathrm{cert}_q(x)$.

Proof. An instance agreeing with the completed transcript produces the same first answer, hence the same next query. By induction it reproduces every query and answer and the same stopping decision, and correctness then gives the same task value. Removing repeated observations can only shrink the transcript. □

An elementary capacity bound shows the role of task-relevant alternatives directly. Suppose $W \subseteq \mathcal{C}$ consists of M instances with $q(y) \neq q(x)$, and every observation disagrees with x on at most r members of W. Every finite certificate T must exclude all of W; hence

$$M \leq r\,|T|.$$

The proof is a union bound over the alternatives each observation excludes. For $r > 0$ it gives $\mathrm{cert}_q(x) \geq M/r$, infinite certificate complexity included. A large M on its own proves nothing when one observation can exclude all M alternatives; the capacity of the permitted operation is essential.

3.2 Counting and decision can require different distinctions

Let $X = \{0,1\}^n$, $n \geq 1$, and take as admissible instances the Boolean functions represented by CNFs of width at most two. Observations reveal the function value at one assignment, and the tasks are satisfiability and exact count. The alternative formulas used in the proof below have at most $2n$ clauses.

Proposition 3.3 (Point-access separation on small CNFs). *For every $a \in X$, the function with satisfying set $\{a\}$ has a one-observation satisfiability certificate and an exact-count certificate of size 2^n in this model. All 2^n point observations suffice for counting any admissible instance.*

Proof. A positive answer at a certifies satisfiability. The empty transcript certifies nothing, since the identically false function is admissible.

For counting, the point a must be observed to separate $\{a\}$ from the empty satisfying set, and every other point b must be observed to separate $\{a\}$ from $\{a, b\}$. Both alternatives lie in the stated CNF class: a singleton is specified by unit clauses, and the false function by two contradictory units. To specify $\{a, b\}$ when $a \neq b$, fix all coordinates on which they agree, choose a coordinate on which they differ, and require each other differing coordinate to follow it with the appropriate sign. Each such equivalence uses two width-two clauses, so the total is at most $2n$. Thus all points are necessary, and observing all of them is sufficient. □

The example isolates a difference between tasks; it is no hard family for 2-CNF formulas supplied in full. Under point access, counting cannot stop at a satisfying witness, because an unobserved assignment may change the count. A written clause list gives different information and supports different operations. Indeed, the two-point formulas in the proof expose their small satisfying sets on sight.

For an L-bit input given in full, reading the L bits reconstructs the entire instance. Every task then has an acquisition procedure using at most L bit observations, once subsequent internal computation goes uncharged. We formalize this operational statement as well. It rules out using a point-access information bound as a running-time lower bound for arbitrary computation on a formula given in full.

3.3 A sufficient condition for transfer between representations

Suppose a representation map $R : \mathcal{C} \to \mathcal{C}'$ preserves admissibility and the task: $q'(R(x)) = q(x)$. For each new observation e', suppose a specified finite set $S(e') \subseteq E$ and a decoder satisfy

$$o'_{e'}(R(x)) = d_{e'}\big((o_e(x))_{e \in S(e')}\big) \qquad (x \in \mathcal{C}).$$

The sets are fixed for each e', so this is a finite, nonadaptive simulation of an individual observation.

Proposition 3.4 (Observation simulation). *Assume the new observation family is finite. If $K > 0$ and*

$$\sum_{e \in S(e')} c_e \leq K c'_{e'} \quad \textit{for every } e',$$

then the weighted certificate infima satisfy

$$\operatorname{cert}_{q,c}(x) \leq K \operatorname{cert}_{q',c'}(R(x)).$$

Proof. Replace each observation in a new certificate T' by $S(e')$. Agreement on their union gives equal decoded answers, so task preservation transports the certificate to x. Nonnegative costs bound the union's cost by the sum of its parts, at most $K \sum_{e' \in T'} c'_{e'}$. Since the new observation family is finite, an attaining certificate exists whenever any certificate does, which gives the infimum comparison. A missing new certificate gives the value ∞. □

The proposition compares acquisition costs. Its decoder is a mathematical function, so it carries no efficiency guarantee. Representation construction, decoding, arithmetic, query selection and the reuse

of intermediate results each need their own running-time analysis. In particular, a missing simulation leaves a faster algorithm unrefuted; it only means that this transfer argument is unavailable.

4 What the algebraic algorithms compute

4.1 Recovery from polynomial data

In the rational interpolation route of [13, Sections 2.4–2.7], a bounded-occurrence formula is encoded by a univariate polynomial H with

$$H(0) = \#F, \qquad \deg H \le MR \le 3nR.$$

Here M is a total-incidence budget and R controls the fixing polynomial. The algorithm computes initial Taylor coefficients at Q interpolation points and recovers the desired value from them. For arbitrary bounded-degree polynomials, the information threshold behind that recovery is exact.

Proposition 4.1 (Ambient Hermite threshold). *Let $t_1, \dots, t_Q$ be distinct nonzero rational numbers, let $J_i \in \mathbb{N}$, and let $d \in \mathbb{N}$. Among rational polynomials of degree at most d, the coefficients of $(X - t_i)^k$ for integers $0 \le k < J_i$ determine the value at zero for every polynomial if and only if*

$$\sum_{i=1}^{Q} J_i > d.$$

Proof. Equality of the jets makes the polynomial difference divisible by

$$P(X) = \prod_{i=1}^{Q} (X - t_i)^{J_i}.$$

Distinct nodes give coprime factors. If $\deg P > d$, divisibility forces the difference to vanish. If $\deg P \le d$, then H and $H + P$ have the same requested jets and stay within the degree bound, yet they differ at zero because $P(0) \neq 0$. □

Here the ambient class matters, because adding P need not produce another polynomial generated by a CNF encoding. Its necessity direction therefore leaves open whether the threshold is optimal for that restricted family, or for every way of recovering its count. For $Q \ge 1$, sufficiency does justify the standard choice

$$q = \left\lfloor \frac{3nR}{Q} \right\rfloor, \qquad J = q + 1, \qquad QJ > 3nR.$$

The computational question is how the jets are obtained. In the cited construction, shared integer measurements separate variable groups. Taylor coefficients of order at most q then reduce to small-support substitutions and group sums, and the complete substitution family has size

$$A(n, q) = \sum_{j=0}^{\min(q,n)} 2^j \binom{n}{j}.$$

In the source's accounting this factor multiplies the enumeration of measurement values and of assignments inside a group, giving the representative product

$$A(n, q)\,(2D + 1)^{2\tau n}\, 2^{2n/r},$$

up to further polynomial and implementation factors [13]. To evaluate the finite-difference identity, the routine visits every pair $U \subseteq E \subseteq [n]$ with $|E| \le q$. Here U fixes the Taylor correction, and E fixes which assignment coordinates must equal one in the inner sum. For each pair the routine scans shared measurement tuples and forms products of local group sums; one correction with one measurement tuple can touch several groups [13, Sections 3.3–3.5]. Faster interpolation of the resulting jets would therefore leave this loop in place. Avoiding it means computing the signed aggregate by some other procedure, and the dependence across groups still allows reuse, dynamic programming or a different factorization.

4.2 Exact network identities and their implementation

The Claude algorithm builds a network with a bit for each literal occurrence, clause factors, and equality factors linking the occurrences of one variable. Its network sum is the exact count. An exact crossing identity replaces a crossing by a signed sum of pieces with a simpler row or column direction, and grouping and recursion then compute the resulting sums [7, Sections 3–6].

Its abstract network model charges for factor and membership evaluation and bounds integer magnitudes. In the implementation, a grouped table query is computed by summing over the grouped local assignments rather than read from a materialized table of free answers. Recursive calls receive evaluator procedures, and the stronger variable bound also uses truncated inclusion–exclusion and deterministic restrictions [7, Appendices B–D].

This is a second way to resolve interactions jointly. It leaves the number of assignments the answer concerns where it was; what it reduces is the work of combining their contributions. Proposition 3.4 therefore gives no identification of this network computation with point observations, and the swap hypotheses of Section 5 have not been derived for it.

Brito's subsequent analysis gives a related example of a construction-specific restriction [4]. For the relevant graded shape, rank subadditivity bounds the number of crossing directions that its pieces can supply. In the application using all basis labels, the crossing has rank $(\dim V_h)^2$, while the directional pieces have total rank at most $(4h+1)\dim V_{\le 4h}$. This growth constraint concerns that shape and its dimension accounting rather than counting in general. Our result below uses different operators, and it proves their rank identity exactly.

4.3 Related method-specific obstructions

Dias and Lima study clause-product interpolation with restored occurrence polynomials, arbitrary nonzero rational evaluation points, and possibly unequal Taylor orders [8, Sections 5–9]. In their setting, recovery must be exact for every polynomial up to the declared degree bound. Their evaluation model charges for group size, interaction width and Taylor order, with a further measurement-rank charge for product-type evaluation. A vanishing-product argument ties these parameters to the input's graph structure, which yields a treewidth-scale obstruction and, on specified random formulas, a $2^{\Omega(\sqrt{n})}$ model-cost bound. Their stated charges exclude some alternative computations inside groups, and their scope discussion explains why the sharper parity and tensor routes escape the model. Theorem 5.1 instead fixes one swap operator family together with its dimension budget. Its larger exponent concerns a more specific enumeration contract, so the exponents alone decide nothing about which result is the stronger one.

Majdoddin studies an arithmetic-cost class of rewrite-and-merge algorithms for weighted clique sums [10, Definition 4.2 and Section 4]. His model evaluates every product choice and charges materialized tables and recursive calls. Its balanced homogeneous two-merge subclass has a rate floor of $(\log_2 K)^{-3}$ under the stated base-method condition. With network size $S = \Omega(n)$ and K bounded by a

polynomially bounded clause count, this excludes fixed-power improvements. Stronger grid bounds and separate single-block statements carry their own hypotheses. These are public near-exponential-scale results for specified network recursions. They supply none of the fixing, row or dimension conditions of our swap theorem, and our arbitrary-measurement quantifier in turn leaves that recursive algorithm class uncovered.

Tulsyan gives the closest comparison [16, Section 5; 17, Sections 1–3]. His supplementary note records the defect-rank identity for the commutative swap algebra, and a triangular-submatrix argument then bounds defect rank for a broader support-constrained matrix class, noncommuting operators included. His paper states an asymptotic near-exponential template-cost consequence on linked bounded-occurrence formulas, using the Arora–Rao–Vazirani structure theorem, the specified Taylor enumeration and a vanishing measurement rate. Its full-label tail argument and its linked-input argument rest on different assumptions. These claims already place the near-exponential scale in this line of work. Our finite bound below reaches it by a direct parity-agreement argument, an explicit dimension budget and the complete clipped sum, so it needs neither a linked-input hypothesis nor a geometric structure theorem. We formalize the entire implication together with the measurement map, rather than take the external template proposition as a premise.

The comparisons are summarized by the quantity each result constrains:

Result	Permitted mechanism	Bounded quantity
Dias–Lima	Clause-product schemes with declared evaluation charges	Model evaluation cost
Majdoddin	Restricted recursive rewrites and merges	Arithmetic-cost exponent
Tulsyan	Support-constrained template matrices and specified input conditions	Defect rank and asymptotic template cost
Theorem 5.1	Full swap history, univariate fixing, row and dimension budgets	Complete clipped enumeration cardinality

An implication between these models needs a construction that preserves their hypotheses and charges. In particular, the freedom to choose any rational measurement map for a fixed operator yields a bound for that operator alone. It yields none for every polynomial deformation, every network recursion or the general point-dependent layout interface.

5 A near-exponential tradeoff for swap-based fixing

5.1 Construction and theorem

Fix integers $r \geq 2$ and $Q \geq 1$. Let

$$\mathcal{H} = ([r] \times [r])^Q, \qquad V = \mathbb{Q}^{\mathcal{H}}, \qquad D = \dim V = r^{2Q}.$$

For $s \in [Q]$, let Π_s exchange the two labels at position s. These commuting involutions act on the coordinate space V itself. We define

$$V_0 = \prod_{s=1}^{Q} \Pi_s, \qquad W_0 = \frac{1}{2}\sum_{s=1}^{Q}(I - \Pi_s), \qquad M_f = V_0 f(W_0)$$

for $f \in \mathbb{Q}[X]$.

Rational linear measurements $C : V \to \mathbb{Q}^m$ enter the construction through one requirement, that a vector with zero measurements is fixed by M_f:

$$\ker C \subseteq \ker(I - M_f). \tag{5.1}$$

This is a property for an application to check. It admits arbitrary rational measurement rows, in any basis.

Theorem 5.1 (Measurement, degree, and enumeration). *Let r, Q, R, m, n be nonnegative integers with $r \geq 2$, $Q \geq 1$, and $R \geq 1$. Let $f \in \mathbb{Q}[X]$ have degree less than R, let $\tau \in \mathbb{R}$ satisfy $0 \leq \tau \leq 1$, and let $C : V \to \mathbb{Q}^m$ be $\mathbb{Q}$-linear and satisfy (5.1) and*

$$m \leq \frac{D\tau}{3r}. \tag{5.2}$$

Then

$$R \geq 1 + \frac{5\sqrt{Q}}{24}. \tag{5.3}$$

If also $n \geq 2^{16}$ and a real dimension budget M satisfies

$$12D^2 \leq M \leq 3n, \tag{5.4}$$

then $Q \leq \frac{1}{4}\log_2 n$, and, with $q = \lfloor 3nR/Q \rfloor$,

$$A(n, q) \geq 2^{n \log_2 \log_2 n/(4\sqrt{\log_2 n})}. \tag{5.5}$$

For every fixed real $\varepsilon > 0$, there is a threshold N_ε, independent of $r, Q, R, m, f, \tau, M, C$, such that every choice satisfying these contracts with $n \geq N_\varepsilon$ obeys

$$2^{n^{1-\varepsilon}} \leq A(n, q) \leq 3^n. \tag{5.6}$$

Here the quantification runs over the operators and measurement maps themselves. Neither a spectrum nor a degree lower bound enters as a hypothesis, since the proof establishes both. Its last statement is uniform over the whole family, and it charges an implementation that visits every member of the enumeration at least once. A different computation of the resulting sum is a different question.

5.2 Exact spectrum and the measurement requirement

On a single pair space, use the basis consisting of the diagonal vectors e_{aa}, the symmetric vectors $e_{ab} + e_{ba}$ for $a < b$, and the antisymmetric vectors $e_{ab} - e_{ba}$ for $a < b$, so that the swap has positive and negative eigenspace dimensions

$$s = \frac{r(r+1)}{2}, \qquad a = \frac{r(r-1)}{2}.$$

Tensoring these bases produces a basis of V. A tensor with exactly j antisymmetric factors has W_0-eigenvalue j and V_0-eigenvalue $(-1)^j$, with multiplicity $\binom{Q}{j} a^j s^{Q-j}$. Consequently,

$$\operatorname{rank}(I - M_f) = \sum_{\substack{0 \leq j \leq Q \\ f(j) \neq (-1)^j}} \binom{Q}{j} a^j s^{Q-j} = D \Pr[f(W) \neq (-1)^W], \tag{5.7}$$

where

$$W \sim \operatorname{Bin}(Q, p), \qquad p = \frac{r-1}{2r} \in [1/4, 1/2].$$

The commutative-algebra reading of (5.7) appears in [17, Section 1.2]. Here the rational pair and tensor bases, the multiplicities and the operator identity are all established in the formal development.

Kernel containment (5.1) and rank–nullity imply

$$\operatorname{rank}(I - M_f) \leq \operatorname{rank} C \leq m,$$

so together with (5.2) the exact identity (5.7) gives

$$\Pr[f(W) \neq (-1)^W] \leq \frac{\tau}{3r} \leq \frac{1}{6}. \tag{5.8}$$

In the source construction, selecting every eigendirection outside an interpolation window is enough to enforce the fixing condition. That selected count may exceed the true defect rank, since the polynomial may happen to agree with parity elsewhere. Equation (5.7) counts the defect itself, so the argument goes through without equating the two quantities.

5.3 Why a low measurement budget forces degree

If f is neither constant 1 nor constant -1, its agreement set with parity on $\{0, \ldots, Q\}$ has size at most $2(R-1)$. Each agreeing point is a root of $f-1$ or of $f+1$, and each of those polynomials has at most $R-1$ roots.

For $p \in [1/4, 1/2]$ and $Q \geq 1$, the elementary binomial estimate proved in Appendix A gives

$$\max_j \Pr[W = j] \leq \frac{2}{\sqrt{Q}}.$$

Hence

$$\Pr[f(W) = (-1)^W] \leq \frac{4(R-1)}{\sqrt{Q}}.$$

Combining this with (5.8) gives $5/6 \leq 4(R-1)/\sqrt{Q}$, which proves (5.3).

Both constant cases need their own argument, and the binomial identity $\mathbb{E}[(-1)^W] = (1-2p)^Q$ supplies it:

$$\Pr[W \text{ odd}] = \frac{1-(1-2p)^Q}{2} \geq \frac{1}{4}, \qquad \Pr[W \text{ even}] \geq \frac{1}{2},$$

so each constant sign violates (5.8), and the zero polynomial is already covered by the nonconstant-sign argument.

Agreement of low-degree polynomials with parity is a classical subject [2]. Our $\sqrt{Q}$ scale matches interpolation on a central binomial window, as used in constructions of probabilistic polynomials [1], and sharp uniform atom bounds are known [3]. We use an elementary and deliberately loose constant because it gives a completely checked implication from the measurement contract as stated. We claim no new general parity lower bound.

5.4 From degree to the complete enumeration

Set $L_2 = \log_2 n$. Since $D = r^{2Q} \geq 2^{2Q}$, (5.4) implies $2^{4Q} \leq n$, so $Q \leq L_2/4$. Combining this with (5.3) and the floor convention gives

$$q \geq \frac{5n}{4\sqrt{L_2}} - 1.$$

For $n \geq 2^{16}$, put $k = \lfloor n/\sqrt{L_2} \rfloor$. Elementary estimates give

$$1 \leq k \leq \min(q, n), \qquad \frac{n}{2\sqrt{L_2}} \leq k \leq \frac{n}{\sqrt{L_2}}.$$

Keeping just the kth term of the complete sum and using $\binom{n}{k} \geq (n/k)^k$ yields

$$A(n,q) \geq \binom{n}{k} \geq \left(\sqrt{L_2}\right)^{n/(2\sqrt{L_2})} = 2^{\,n \log_2 L_2/(4\sqrt{L_2})},$$

which is (5.5). Clipping at n matters here, since the argument still applies when $q > n$. By the binomial theorem, the complete sum is always at most $\sum_{j=0}^{n} 2^j \binom{n}{j} = 3^n$.

Finally, for every fixed $\varepsilon > 0$ the exponent in (5.5) eventually exceeds $n^{1-\varepsilon}$. Indeed $4 \log_2 n \le n^{\varepsilon}$ eventually, and once $L_2 \ge 16$ we have $\sqrt{L_2} \le L_2$ and $\log_2 L_2 \ge 1$, which together give the comparison. Since that threshold depends on ε alone, (5.6) follows. □

5.5 The source measurement map

For the measurements the source itself defines, the fixing condition can be discharged outright. Let $(b_\alpha)_{\alpha \in \mathcal{E}}$ be the orthogonal tensor basis from Section 5.2, and let $\nu(\alpha)$ count its antisymmetric factors. For a retained set $J \subseteq \{0, \dots, Q\}$, put

$$\mathcal{S}_J = \{\alpha : \nu(\alpha) \notin J\}, \qquad m_J = |\mathcal{S}_J|,$$

and define the row map by

$$C_J : V \longrightarrow \mathbb{Q}^{\mathcal{S}_J}, \qquad (C_J v)_\alpha = \sum_{u \in \mathcal{H}} b_\alpha(u) v(u).$$

Ordering the selected rows identifies its codomain with $\mathbb{Q}^{m_J}$.

Proposition 5.2 (Selected spectral measurements). *Suppose $f(j) = (-1)^j$ for every $j \in J$. Then $C_J v = 0$ implies $M_f v = v$. Every matrix entry of C_J is 0, 1 or -1, every row has ℓ_1-norm at most D, and*

$$m_J = \sum_{j \notin J} \binom{Q}{j} a^j s^{Q-j} = D \Pr[W \notin J],$$

where the sum ranges over $0 \le j \le Q$, and a, s, W are as in Section 5.2. Consequently, the tail condition $\Pr[W \notin J] \le \tau/(3r)$ supplies the row budget in Theorem 5.1 for this constructed map, with no fixing-kernel premise left to assume.

Proof. Expand $v = \sum_\alpha \beta_\alpha b_\alpha$. Orthogonality gives $(C_J v)_\alpha = \|b_\alpha\|^2 \beta_\alpha$. Each squared norm is positive, so zero measurements force every selected coefficient to vanish. Every remaining basis vector has $\nu(\alpha) \in J$ and M_f-eigenvalue $(-1)^{\nu(\alpha)} f(\nu(\alpha)) = 1$, which proves the fixing property. Basis entries lie in $\{0, 1, -1\}$, giving the row-height bound. Counting selected indices by their antisymmetric factors gives the exact multiplicities and the binomial normalization. □

These are precisely the transposed eigenvector rows in the proof of OpenAI's auxiliary-matrix lemma [13, Lemma `lem:swaps-matrix`]. In the source, J is a consecutive window and f its degree-less-than-$|J|$ parity interpolant. Our formal artifact constructs such an interpolant, proves its values, degree and uniqueness, and composes it with the selected-row map and Theorem 5.1; an integer window is handled by retaining its intersection with the possible eigenlevels. Where the source writes its interpolant with a Newton binomial formula, the formal application uses the unique interpolation characterization instead.

This local map $C_J = C^{\mathcal{B}}$ is distinct from the formula's placed measurements, which the source obtains by taking h block copies, pulling them back along the variable-repetition map ι, and restricting columns to each group:

$$\widetilde{C} = \operatorname{diag}(C_J, \dots, C_J), \qquad \widehat{C} = \widetilde{C}\iota, \qquad C_{t,g} = \widehat{C}|_{S_{t,g}}.$$

Zero rows are retained. In the bounded-occurrence clause branch, the source has $hD \le 2M$, incidence and scope bounds $b = w = 3$, and $12D^2 \le M \le 3n$. The local tail budget therefore gives the global

row accounting

$$rh\, m_J \le \frac{hD\tau}{3} \le \frac{2M\tau}{3} \le 2\tau n,$$

and the artifact checks this numerical implication too. Since the placement construction itself comes from the source, its global row bound on its own would leave the local tail budget underived. In the source, $0 < \tau \le 1/2$; the theorem also admits the nonnegative boundary.

Its sparse polynomial equation branch [13, Sections 4.7 and 5.2–5.5] uses the same local swap operator and measurement rows over $\mathbb{Q}$. On a nontrivial branch with n' free variables, the layout instead requires $2(b+w)D^2 \le M \le bn'$ and $2Qr^2 \le n'$, with b and w bounded by powers of $\log n'$, and its local budget is $m_J \le D\tau/(rb)$, where $\tau = 1/(r\lambda)$ and $\lambda = \lceil \log_2 n' \rceil$. Thus $\tau_* = 3\tau/b \le 1$ for sufficiently large n', and (5.2) holds with τ_* in place of τ, so the same degree floor (5.3) follows.

For the enumeration step this branch has its own cutoff $q = \lfloor dR/Q \rfloor$, where the source takes $d = k(m_{\text{test}} + 2(\ell - 1))$ for degree bound $k \ge 1$, m_{test} equations and precision $\ell = n + 1$ on the original $n \ge n'$ variables, so $d \ge 2n'$. With the source's choice $Q = \Theta(\log n' / \log\log n')$, (5.3) gives $q = \Omega(n'/\sqrt{Q})$, hence eventually $q \ge \lfloor n'/\sqrt{\log_2 n'} \rfloor$. Keeping that term as in Section 5.4 yields $\log_2 A(n', q) = \Omega(n' \log\log n'/\sqrt{\log n'})$. So the near-exponential obstruction applies to the source's complete enumeration in this branch as well, with its own cutoff and budgets; this derivation uses the checked degree bound and the source's parameter choices.

5.6 What the construction-specific obstruction establishes

What the theorem closes is one escape route. Within the full history space, with a single univariate fixing polynomial, a measurement map satisfying (5.1), the row and dimension budgets, and the complete substitution enumeration, no change of f or of the rows of C brings this enumeration down to $2^{O(n^{1-\varepsilon})}$ for any fixed $\varepsilon > 0$. To absorb the constant hidden in that O-notation, apply (5.6) with $\varepsilon/2$.

This chain also shows what pays for the residual interaction. Few measurements leave a large subspace that the operator must fix. Fixing that subspace requires agreement with parity on substantial binomial mass, the agreement requirement forces degree, and degree forces a large substitution family under the dimension budget. So this is a proved relation among distinguishable directions, joint algebraic operations, and the cost of one recovery procedure.

Proposition 5.2 supplies the local measurement application, and Section 5.5 accounts for the distinct budgets of the clause and sparse polynomial equation branches of [13]. Everything around them keeps its source-level analysis: the formula reduction, the schedule and placement construction, the replacement-form support, the coefficient bounds and the bit costs. In particular, the present formalization covers neither the full counting algorithm nor a transfer of the same contracts to [7]. A new representation, a multivariate fixing operation, a smaller invariant space or a recovery algorithm that avoids complete enumeration would each call for its own analysis.

6 A refined hardness program

Within its specified family, Theorem 5.1 shows the same fixed-power separation that Weak #ETH asserts. We read that agreement as evidence about a mechanism, and nothing more: it establishes no unrestricted hypothesis. Its value for the broader question lies in the shape of the proof. Its obstruction is attached to a concrete interaction that a permitted computation must resolve. That obstruction survives every change the model already allows. Proposition 5.2 ties the statement to the local measurements the source uses.

A general structural account would have to justify comparable statements for a far wider class of computations. Three requirements are unavoidable. First, task-changing alternatives must be genuine admissible inputs rather than objects in some convenient ambient space; Proposition 4.1 shows why. Second, any transfer across representations must account for the information supplied and for its acquisition, instead of treating an aggregate operation as a point query; Proposition 3.4 gives one sufficient condition, and no classification. Third, computation must be charged at the level of evaluation and recovery as performed. That level includes construction, arithmetic precision, reuse, and the selection of intermediate objects. The network algorithms show why certificate size alone cannot supply these costs.

This leaves a concrete way to investigate the successor hypothesis. For a proposed fixed-power improvement, locate the joint operation that produces the gain. Identify the interaction it leaves unresolved. Then prove a tradeoff that includes the operation's full contract. Section 4.3 illustrates three realizations of this: graph structure and recovery exactness, rewrite-and-merge geometry, and support-constrained rank. Our finite spectral analysis adds a checked instance with a source-defined measurement map. An obstruction may concern rank, degree, support, representation size or the cost of a remaining sum. A new algorithm may instead show that the proposed obstruction was tied to too narrow a family. Either way the structural claim becomes more precise.

This program is more modest than the slogan that harder problems require more work. It is no monotonicity theorem in one syntactic size parameter. Easy instances may have long descriptions. Short descriptions may encode expensive tasks. The relevant distinctions are relative to the requested answer and to the operations available for computing it. Nor does the program hold that every change of representation preserves hardness. The reported ETH algorithms matter precisely because a different organization can lower computational cost.

Our quantitative destination remains #SH rather than the original ETH boundary. Public reduction theory [11] already explains how fixed-power lower bounds propagate when reduction costs and size blowups are controlled. The complementary task pursued here is to explain and prove the internal obstruction for one counting mechanism. Extending such arguments to arbitrary exact-counting algorithms is open. What the present results supply is an independently stated, formally verified case in which the refined near-exponential scale persists.

7 Formalization and reproducibility

Our Lean 4 development uses Mathlib [12, 15]. Its public entry point is `ETHResponse.lean`. Its selected import closure contains the task-certificate and transcript results, the width-two CNF construction, operational bit reading, finite observation simulation, rational Hermite recovery, the swap basis and spectrum, the measurement-rank implication, and both the finite and the eventual enumeration bounds.

In the formalization, the swap operator acts on rational functions on the history product itself. The spectral identity with its binomial multiplicities is proved from a rational basis rather than assumed numerically. Its general measurement endpoint quantifies over a rational linear map satisfying the fixing-kernel condition. An application module constructs the transpose rows the source selects. It proves their relation to dual coordinates, their entry and height bounds, their row count and their kernel property. It composes them with the finite and eventual bounds. Separate interpolation and window results discharge the parity-agreement premise for the constructed interpolant. Global placement matrices lie outside this formal slice.

`ETHResponse/Audit.lean` checks the transitive axiom dependencies of every loaded declaration in the project's mathematical namespaces, private helpers included. It permits only propositional

extensionality, classical choice and quotient soundness. An export records source hashes, pinned dependency revisions, build results and the freshly measured declaration count. `CORRESPONDENCE.md` maps the paper's mathematical statements to named theorem endpoints and their application hypotheses.

So the formalization establishes the mathematical implications stated here, and those alone. #SH itself, its validity, an end-to-end verification of either external counting algorithm, and a general time lower bound from the point-access example all lie outside it. Each accompanying TeX source archive ships the selected Lean project in `anc/lean_proofs/`, with its correspondence document, toolchain and dependency lockfile. `SOURCE_MANIFEST.json` identifies the proof snapshot by file hashes. The build and axiom receipts bind their results to that snapshot. A combined editable package holds both manuscripts, the public bibliography and the rebuild tools, whose instructions regenerate the PDFs and source packages from that layout.

A An elementary binomial atom bound

We include the estimate used in Section 5 to make the numerical degree constant independent of an imported asymptotic approximation. Let $b_{m,j} = 2^{-m}\binom{m}{j}$ be the fair-coin mass. The central-binomial recurrence gives

$$b_{m,j}^2 \le \frac{1}{m+1} \quad \text{for every } m, j \ge 0.$$

Put $c_k = 4^{-k}\binom{2k}{k}$. The recurrence $(2k+2)c_{k+1} = (2k+1)c_k$ gives $(2k+1)c_k^2 \le 1$ by induction. The maximum mass is c_k in row $2k$ and c_{k+1} in row $2k+1$, proving the bound.

For $0 < p \le 1/2$, let $N \sim \mathrm{Bin}(Q, 2p)$. Thinning by independent fair coins gives

$$\Pr[\mathrm{Bin}(Q,p) = j] = \mathbb{E}[b_{N,j}].$$

Cauchy–Schwarz and the fair-coin bound imply

$$\Pr[\mathrm{Bin}(Q,p) = j]^2 \le \mathbb{E}\left[\frac{1}{N+1}\right] \le \frac{1}{2p(Q+1)}.$$

For $p \ge 1/4$ and $Q \ge 1$, this is at most $2/(Q+1)$ and hence yields the convenient bound $2/\sqrt{Q}$ used above. The Lean proof establishes the finite mixture identity algebraically and proves the inverse-moment upper bound; no probabilistic limit theorem is required.

B Principal formal endpoints

The table gives short names; each theorem is in the `ExactSolution` namespace. The source correspondence file includes the supporting modules and exact application contracts.

Mathematical statement	Lean theorem
Completed transcript is a certificate	`transcript_certificate`
Transcript query lower bound	`transcript_query_lower_bound`
Width-two CNF count separation	`SingletonCNF.singleton_count_complexity`
Explicit-input acquisition ceiling	`read_all_operational`
Finite weighted observation simulation	`weighted_observation_simulation`
Ambient Hermite threshold	`hermite_value_at_zero_determined_iff`
Actual swap defect rank	`SwapSpectrum.swap_rank_eq_failure_mass`

Mathematical statement	Lean theorem
Measurement-to-degree implication	`MeasurementRank.swap_measurement_degree_floor`
Finite clipped-enumeration floor	`MeasurementRank.swap_measurement_enumeration_floor`
Uniform eventual near-exponential floor	`swap_measurement_near_exponential`
Complete-enumeration upper bound	`enumerationCount_le_three_pow`
Constructed physical measurement kernel	`SelectedMeasurements.measurement_fixing_kernel`
Integer measurement-row height	`SelectedMeasurements.integerMeasurementMatrix_l1_le`
Exact selected-row tail count	`SelectedMeasurements.rowCount_eq_dimension_mul_tail`
Interpolant uniqueness	`SelectedMeasurements.eq_parityInterpolant`
Constructed interval enumeration	`SelectedMeasurements.interval_measurement_enumeration_floor`
Uniform interval application	`SelectedMeasurements.interval_measurement_near_exponential`
Placed-row numerical accounting	`SelectedMeasurements.placed_row_count_bound`

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